Praneeth Vedantham

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Education

Rutgers University - New Brunswick May 2026
Bachelor of Science in Computer Science, Minor in Mathematics *New Brunswick, NJ*

- **Coursework**: Data Structures; Algorithms; Computer Architecture; Systems Programming; Intro to AI; Linear Optimization

Experience

AI Research Intern May 2025 – August 2025
App Orchid *Remote*

- Spent the first few weeks digging into several of the MSOs (managed semantic objects) behind Easy Answers, each with 8–10 properties, to understand how the ontology turns a plain-English question into a SQL query against the right tables and columns.
- Wrote a small Python client over the AO Platform's REST APIs that handles token refresh, retries, and error logging, then used it to script the setup of domains, ontologies, data sources, and solutions that the team used to click through by hand in the console.
- Validated the AI-generated SQL behind the "Easy Answers" feature by working through a spreadsheet of expected queries for 100+ question/answer pairs, diffing them against what the platform actually generated, and debugging the joins and filters that were off.

Lead Engineer August 2024 – January 2025
Ghosteams *Remote*

- First engineer at a small product development agency. Built the client-facing platform on React, Next.js, and GraphQL, hosted on Vercel. Each client got a live board of their tasks, and status changes posted straight to their Discord channel.
- Wrote the scoping tool the agency ran on every new project. It took a client brief, had an LLM draft the task breakdown and rough estimates, and dropped the result onto the board for someone to fix up before anything went back to the client.
- Worked with 6 engineers on AgentHub, an autonomous agent platform that kept memory across sessions and could post to social media, Discord, and Telegram. Wrote the Python library that coordinated the agents and handled payments between them.

IT Infrastructure & Services Intern June 2024 – August 2024
Pitney Bowes *Shelton, CT*

- Dug into the virtual machine distribution process across 200+ VMs and found roughly $4,000 worth of compute and storage being wasted on machines that were never returned or were sized well beyond what their owners used.
- Wrote Python scripts against the vSphere API to flag VMs that had sat idle for a month or were barely using their allocated CPU/memory. The first pass reclaimed 40+ machines and turned a weekly manual review into an on-demand one.
- Set up form automation in Jira Service Management for employees reporting technical issues. Standardizing the request templates meant tickets came in with the right details up front, which trimmed triage time by about a third.

Director of Web Services April 2023 – June 2024
Phi Chi Theta *New Brunswick, NJ*

- Ran the chapter's primary website for 3 semesters, pushing regular content updates (new members, events, recruitment pages) with prospective members in mind. Recruitment-page traffic went up about 25% during fall rush.
- Built a "Gallery" page in HTML/JavaScript that doubles as the organization's photo repository, pulling together a few hundred photos from past events that were scattered across shared drives and members' personal phones.
- Cleared out a backlog of main-page bugs and layout glitches left over from previous semesters; mobile load time dropped about 40%.

Projects

IncidentHub | *TypeScript, React, Node.js, Fastify, PostgreSQL, Redis, Socket.IO, BullMQ, Docker* | GitHub Summer 2026

- Built a full-stack, real-time incident management platform, basically a lightweight PagerDuty. Teams open incidents, page responders, and escalate automatically when nobody acks. Commands go over REST and updates fan out to every browser over Socket.IO.
- Wrote the Postgres schema by hand (12 tables, 14 indexes), with audit logging in database triggers and a version number on every edit to stop silent overwrites. A partial index fixed the main performance bottleneck, taking the dashboard query from 16.5 ms to 1.5 ms.
- Escalation runs as BullMQ delayed jobs in a separate worker microservice, with deterministic job ids so retries can't double-page.
- Wrote 50+ Vitest and Playwright unit and end-to-end tests that run against real Postgres and Redis in CI.

Datastructures.io | *TypeScript, Cloudflare Workers, Durable Objects, WebSockets, Piston, Docker* | Live site Summer 2026

- Teamed up with two other engineers on the Cloudflare/DigitalOcean stack behind a ranked 1v1 coding platform. Live matches went from 28 to 100+ concurrent at about $17/month via event-driven matchmaking, batched writes, and Durable Object hibernation.
- Owned the anti-cheat: a 2-layer server-authoritative design with replayable event logs, admin review and ban tooling, and Elo reconciliation. Player code in Python, JavaScript, Java, and C++ runs sandboxed on a dedicated x86 Piston judge.

RetryHTTP | *Python, httpx, requests, asyncio, pytest, GitHub Actions* | GitHub Summer 2026

- Pulled the retry logic out of the App Orchid API client and packaged it as a shared library on PyPI. It subclasses `httpx.Client` and `requests.Session` so existing code works unchanged, and adds backoff with jitter, per-status retry rules, and 401 token refresh.
- 26 pytest tests cover the sync, async, and requests paths. CI runs them on Python 3.10–3.12 and publishes tagged releases to PyPI.

Bond Cash-Flow API | *Python, Flask, pandas, NumPy, pytest* | GitHub October 2026

- Flask REST API for bond pricing. Builds the cash flow schedule as a pandas DataFrame, prices from a yield or solves yield to maturity by bisection, and returns duration, convexity, and DV01. 26 pytest tests, drafted with GitHub Copilot, cover every endpoint.

Technical Skills

Languages: Python (Pandas, NumPy), SQL, TypeScript, JavaScript, Java, HTML/CSS
AI / LLM: Generative AI coding assistants (GitHub Copilot, Claude Code); Agents (Claude SDK, OpenAI, MCP); RAG (pgvector); Evals
Tools & Systems: Flask, PostgreSQL, Redis, Docker, AWS, Cloudflare Workers, Git, GitHub Actions, CI/CD, Jira, vSphere